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A study of experimental privacy budget quantification in quantum differential privacy simulations

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ABSTRACT 150-250 palabras

INDEX TERMS Differential privacy, Quantum computing, Quantum noise

I. INTRODUCTION

ESTA va a ser la introduccion a la tesis. En esta sección pienso hablar de la motivación, la importancia de la privacidad diferencial, basicamente muy similar al documento de propuesta

II. BACKGROUND

En esta seccion se va a introducir todos los conceptos, ecuaciones y conocimiento necesario para entender el paper. Siguiendo el documento de propuesta propuesta vamos a tener:

A. DIFFERENTIAL PRIVACY

Las definiciones clasicas de privacidad diferencial

B. QUANTUM INFORMATION AND COMPUTATION

Las definiciones necesarias para entender el paper desde informacion cuantica: (estado, matriz de densidad, ruido cuantico, etc)

C. QUANTUM DIFFERENTIAL PRIVACY

Las definiciones de la privacidad diferencial cuantica y las de los algoritmos del paper de Zing para encontrar epsilon (dPD, PD)

III. RELATED WORK

El estado del arte muy similar a la propuesta donde se establece cual es el entorno de investigacion de la privacidad diferencial cuantica. En especial sobre la privacidad diferencial usando el ruido de decoherencia que es la forma que la estamos aplicando en el paper.

IV. DATA PROCESSING AND ANALYSIS

This section details the data processing and analysis methodology that was used in order to prepare the classical Statlog (German Credit Data) dataset [1] [2] for use in the quantum setting. Firstly we outline the dataset characteristics, variable selection procedures and lastly quantum encoding procedures for the selected data.

A. DATASET DESCRIPTION

The selected dataset for this investigation is the Statlog (German Credit Data) dataset created by Professor Hans Hofmann [1]. This multivariate dataset aims to classify people as a good or bad credit risk. The orginal dataset is composed of 1000 instances that span 21 variables (7 numerical, 13 categorical, and the target).

In the original dataset created by Professor Hofmann, all non numerical variables use labels to define their characteristics (e.g The status of a person whose checking account is less than 0 Deutsche Marks (DM) is A11) for that reason it was decided that the investigation was going to use a curated version of the dataset [2] which transforms all the labels to their definition as a means of simplifying the readers' understanding of the dataset by eliminating that intermediate layer of label and category. Every mention to the dataset henceforth will refer to this curated version and not the original compiled by Professor Hofmann

A simple data analysis concluded that the dataset didn't have any problem in terms of completeness (as all rows include all 21 variables) nor uniqueness (as there is no instance of an exact duplicate). For these reasons no special cleaning procedure was done prior to variable selection.

B. VARIABLE SELECTION WITH PHIK

In order to hold quantum data for a dataset that contains 1000 instances of 20 variables each, the simulation would, at a maximum, store 1000 $2^n \times 2^n$ density matrices, with n being the number of qubits used. This means that using the full 20 variables, one per qubit, would imply $2^{20} \times 2^{20} \approx 1.1 \times 10^{12}$ complex entries, and at 16 bytes each, that is about 16 TiB per matrix. It was decided that a correlation study over the target variable would be an adequate way to reduce this load. The reasons for this decision are: First, the investigation did not aim to produce a classifier and thus could use a smaller number of variables in the experiments. Second, the variables selected from a correlation study would be the most likely candidates to be chosen as significant by a classifier.

As mentioned previously however, most of the variables in this dataset are not numeric and thus a popular correlation analysis such as the Pearson correlation is not an adequate choice to evaluate variable correlation. For starters, for a nominal variable the Pearson's coefficient relies on an arbitrary choice for the label and since the proportions of the target variable in the dataset for each category are not the same across the board, a change in category label implies a change in the coefficient. Moreover the Pearson coefficient identifies well the linear relationships of variables but lacks for nonlinear cases.

For these reasons it was decided to use the PhiK method [3] to select the variables which show a consistent stronger correlation with the target variable. PhiK is a stronger candidate as it works well for both numerical and categorical variables correlations and it is able to identify nonlinear relationships between them.

We used Baak et al. PhiK library to measure the correlation and significance between each variable and the target and recorded the results in Table 1. It is important to note that the library returns, for a pair of variables, two results: the first is ϕ_K which is the measurement of how strong the relationship between the two variables is. This result is calculated by calibrating a χ^2 test statistic to a contingency table (populated with the categorical variables or a bin distributed numerical variables) with respect to a bivariate normal distribution. This calibration returns a ϕ_K with range of 0 (no association) to 1 (complete association). Since ϕ_K depends on the counting of the contingency table renumbering the categories no longer affects the end result. The second result is Z which is a conversion of the p-value from the independence test into a number of standard deviations. A higher Z provides more evidence that the variable is not independent from the target.

For the variable selection we chose to use Z as although the ϕ_K measures the correlation between variables, the significance Z of that correlation explains if that correlation is significant enough to be told apart from chance. The Bonferroni method states that, to ensure that the probability of at least one false positive across k tests (in our case $k = 20$ for 20 variables against the target) is at most $\alpha = 0.05$, each individual test must use a significance level of $\alpha' = \alpha/k = 0.05/20 = 0.0025$ [4]. Here a false positive is a variable that is independent of the target but is selected anyway. Since $\alpha' = 0.0025$ corresponds to $Z \approx 2.81$, we used $Z > 2.81$ as the cutoff for the selection of the variables.

TABLE 1. ϕ_K correlation of each feature with the target and its statistical significance with a cutoff of $Z > 2.81$.

Feature	ϕ_K	Z	Significant
status_account	0.512	10.965	Yes
credit_history	0.197	6.912	Yes
month_duration	0.285	5.648	Yes
status_savings	0.147	5.283	Yes
credit_amount	0.241	4.388	Yes
collateral	0.217	4.009	Yes
purpose	0.204	3.715	Yes
housing	0.077	3.601	Yes
years_employment	0.098	3.053	Yes
other_installment_plans	0.063	2.891	Yes
is_foreign_worker	0.109	2.545	No
age	0.142	2.177	No
status_and_sex	0.123	1.953	No
secondary_obligor	0.041	1.835	No
payment_to_income_ratio	0.075	1.045	No
telephone	0.021	0.689	No
n_credits	0.000	0.043	No
job	0.000	−0.390	No
residence_since	0.000	−1.070	No
n_guarantors	0.000	$-\infty$	No

As the reader can observe in Table 1 the selection took 10 out of the 20 variables which as mentioned before now leaves a maximum of 1000 $2^{10} \times 2^{10}$ matrices as opposed to 1000 $2^{20} \times 2^{20}$ should we use one qubit per variable, we will show how we lowered that number in Section IV-C. The definition and type of each variable is recorded in Appendix A

C. ENCODING OF CLASSICAL VARIABLES INTO QUANTUM

La explicación de la metodologia de codificación de variables categoricas y númericas en qubits y los tradeoff que este proceso conlleva.

D. NOISE PARAMETERS SELECTION FOR EACH NOISE MECHANISM

Un pequeño parrafo que sustente porque se utilizaron los parametros de ruido dado (i.e p = 0.5, gamma = 0.5 etc) esto basicamente hace referencia al paper de Zhou.

V. FINDING THE EXPERIMENTAL PRIVACY BUDGET

A. CENTRAL SETTING

La explicacion del experimento central, es basicamente puesto en texto la presentacion de semana 8

B. LOCAL SETTING

La explicacion del experimento loca, es basicamente puesto en texto la presentacion de semana 10

VI. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

La explicación de la medicion de las variables categoricas como forma de mostrar el ruido en acción. Es el resto de la

presentación de semana 10 puesto en texto

VII. EXPERIMENTAL RESULTS

A. FINDING THE EXPERIMENTAL PRIVACY BUDGET

1) Central Setting

Resultados del experimento central

2) Local Setting

Resultados del experimento local

B. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

Resultados de las mediciones con las distribuciones de probabilidad

VIII. DISCUSSION AND CONCLUSION

En esta seccion voy a hablar de limitaciones encontradas dentro de los experimentos.

Entre ellas:

La dualidad entre C y L para la privacidad. (pros y contras) Que limitaciones tiene el experimento (analisis conjunto)

Tambien el analisis de los resultados de los mismos y que implicaciones tienen estos frente a la privacidad diferencial tanto clasica como cuantica.

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APPENDIX A DESCRIPTION OF SELECTED FEATURES

ACKNOWLEDGMENTS

Esta seccion solo para la tesis o no se si pueda ir en el paper tambien

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TABLE 2. Description of the selected features.

Feature	Type	Description
`status_account`	Ordinal	Status of the existing checking account, in DM (Deutsche Marks)
`month_duration`	Numerical	Duration of the credit in months
`credit_amount`	Numerical	Credit amount
`collateral`	Categorical	Possible collateral for the loan
`purpose`	Categorical	Purpose of the credit
`credit_history`	Categorical	Credit history of the applicant
`housing`	Categorical	Current housing situation (rented, owned, or free)
`years_employment`	Ordinal	Length of time in current employment
`status_savings`	Ordinal	Status of the savings account, in DM (Deutsche Marks)
`other_installment_plans`	Categorical	Whether the client has other installment plans